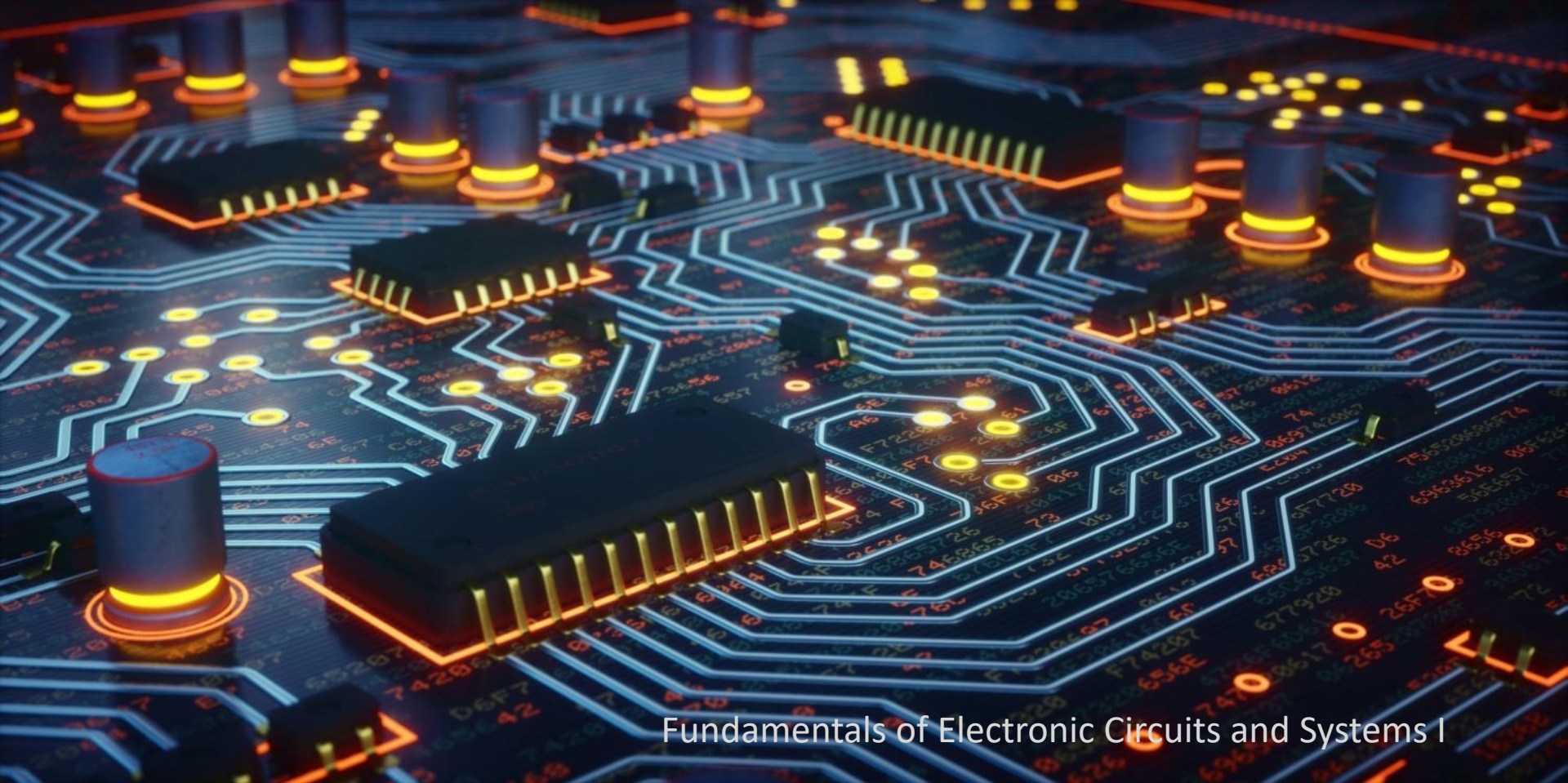




Fundamentals of Electronic Circuits and Systems I

Circuit Analysis Techniques

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Recall: Kirchhoff's Law

- **VOLTAGE LAW:** The rate of change of magnetic flux linked with any portion of the circuit must be zero for all time.

$$\oint E \cdot dl = -\frac{\partial \Phi_B}{\partial t} = 0 \quad \Rightarrow \quad \sum_{n=1}^N v_n = 0$$

KVL: Sum of voltages around a closed path is zero

- **CURRENT LAW:** The rate of change of the charge at any node in the circuit must be zero for all time. A node is any point in the circuit at which two or more element terminals are connected using wires.

$$\int_{S_x} J \cdot dS - \int_{S_y} J \cdot dS = \frac{\partial q}{\partial t} = 0 \quad \Rightarrow \quad \sum_{n=1}^N i_n = 0$$

KCL: Sum of currents entering a node is zero

Recall: How to write circuit equations?

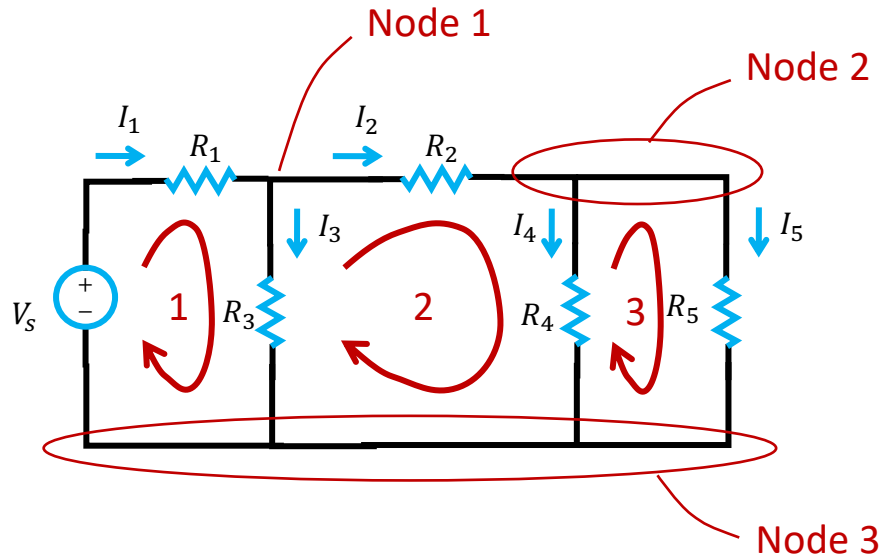
- Step 1: define all the **UNKNOWN** currents & voltages
- Step 2: write equations
 - check **DEVICES**
 - write device equations according to **$i - v$ characteristics**
 - check **NODES**
 - Write current equations according to **KCL**
 - check **LOOPS**
 - Write voltage equations according to **KVL**
- Step 3: solve the equations

Outlines

- Node-voltage method
- Mesh-current method
- Linearity & Superposition
- Thévenin and Norton Equivalent Circuits

Recall: an example from last lecture

QUESTION: Find the output current of the voltage source

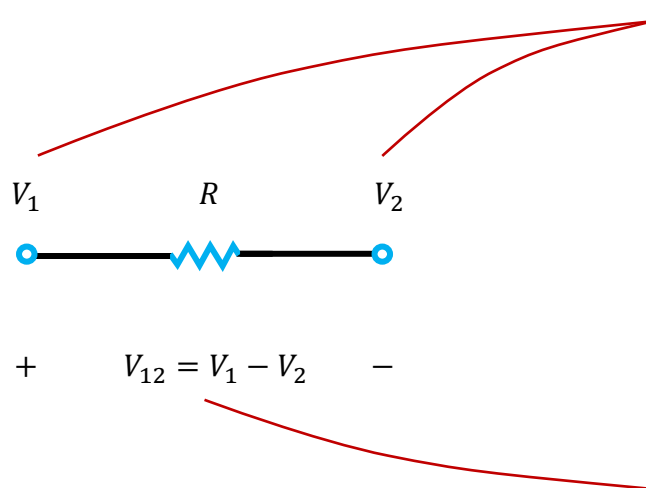


■ According to KVL/KCL/Ohm's law

$$\begin{bmatrix}
 1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & -1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\
 R_s & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
 0 & R_2 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\
 0 & 0 & R_3 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
 0 & 0 & 0 & R_4 & 0 & 0 & 0 & 0 & -1 & 0 \\
 0 & 0 & 0 & 0 & R_L & 0 & 0 & 0 & 0 & -1
 \end{bmatrix}
 \begin{bmatrix}
 I_1 \\
 I_2 \\
 I_3 \\
 I_4 \\
 I_L \\
 V_{R_s} \\
 V_{R_2} \\
 V_{R_3} \\
 V_{R_4} \\
 V_{R_L}
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 0 \\
 1 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0
 \end{bmatrix}
 V_s$$

A lot of equations! A lot of work!

Basis of node voltage



Node voltages

Voltage at the node

Current in terms
of node voltages

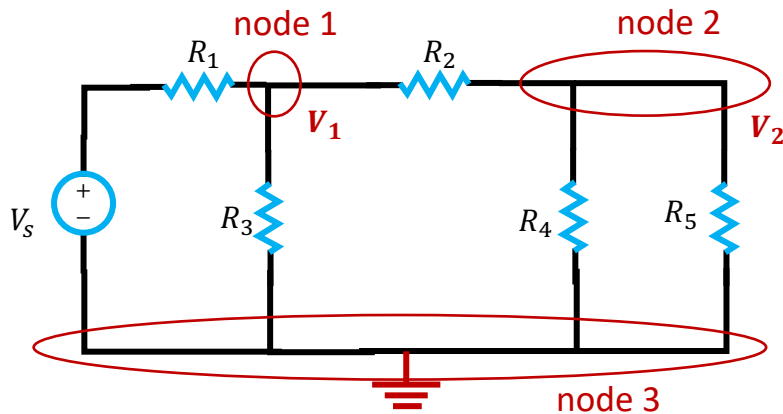
$$I = \frac{V_1 - V_2}{R}$$

Elemental voltage

The voltage across the element

Node-Voltage Method

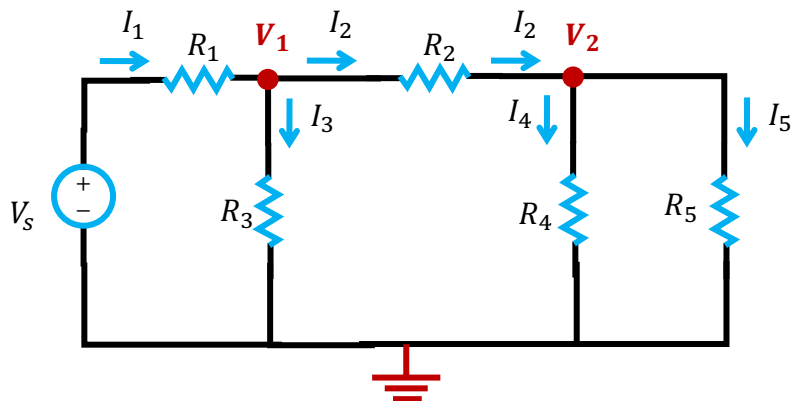
QUESTION: if all the resistor values and V_s are known. Find all voltage & current in circuit



- Step 1a: identify all extraordinary nodes
- Step 1b: Assign one extraordinary node as **ground**
 $\text{node 3} = \text{ground} = 0V$
- Step 1c: assign node voltages to rest of the extraordinary nodes

Node-Voltage Method

QUESTION: if all the resistor values and V_s are known. Find all voltage & current in circuit



- Step 2a: perform **KCL** at each node except the ground node

@node 1

$$I_1 - I_2 - I_3 = 0$$

@node 2

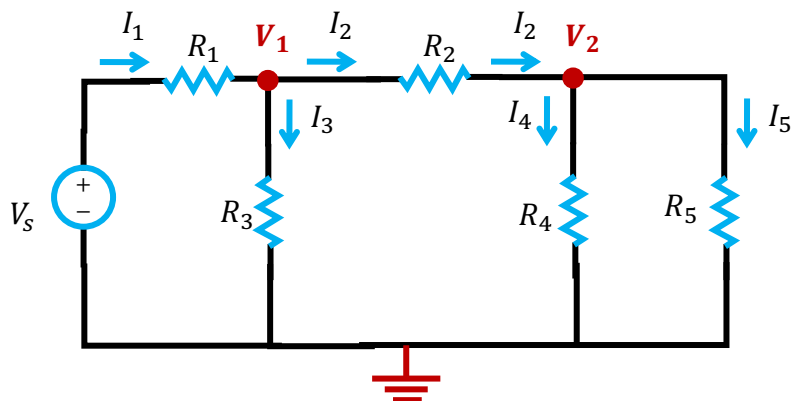
$$I_2 - I_4 - I_5 = 0$$

- Step 1a: identify all extraordinary nodes
- Step 1b: Assign one extraordinary node as **ground**. node 3 = **ground** = **0V**
- Step 1c: assign node voltages to rest of the extraordinary nodes

* Note: Assign a positive sign to current entering a node

Node-Voltage Method

QUESTION: if all the resistor values and V_s are known. Find all voltage & current in circuit



- Step 1a: identify all extraordinary nodes
- Step 1b: Assign one extraordinary node as **ground**. node 3 = **ground** = **0V**
- Step 1c: assign node voltages to rest of the extraordinary nodes

- Step 2a: perform **KCL** at each node except the ground node
- Step 2b: apply ohm's law for each current in terms of node voltages

@node 1

$$I_1 - I_2 - I_3 = 0$$

$$\blacktriangleright \frac{V_s - V_1}{R_1} - \frac{V_1 - V_2}{R_2} - \frac{V_1}{R_3} = 0$$

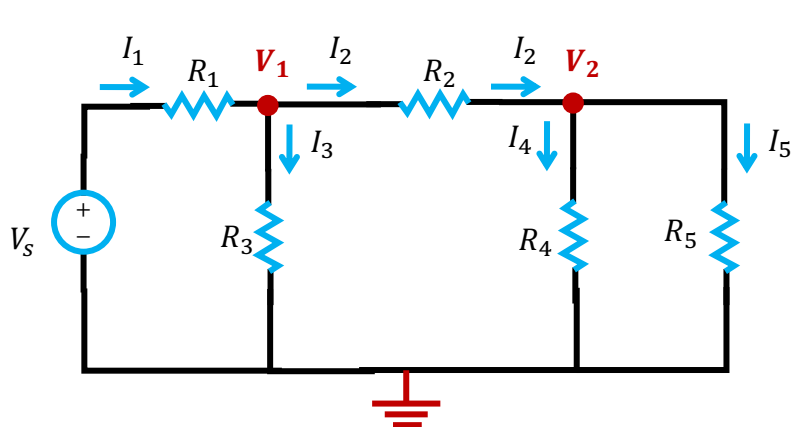
@node 2

$$I_2 - I_4 - I_5 = 0$$

$$\blacktriangleright \frac{V_1 - V_2}{R_2} - \frac{V_2}{R_4} - \frac{V_2}{R_5} = 0$$

Node-Voltage Method

QUESTION: if all the resistor values and V_s are known. Find all voltage & current in circuit



$$\left\{ \begin{array}{l} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) V_1 - \frac{1}{R_2} V_2 = \frac{1}{R_1} V_s \\ \frac{1}{R_2} V_1 - \left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} \right) V_2 = 0 \end{array} \right.$$

- Step 1a: identify all extraordinary nodes
- Step 1b: Assign **ground** node
- Step 1c: assign node voltages to the rest
- Step 2a: perform **KCL** at each node
- Step 2b: apply ohm's law for each current
- Step 3: solve the system of equations

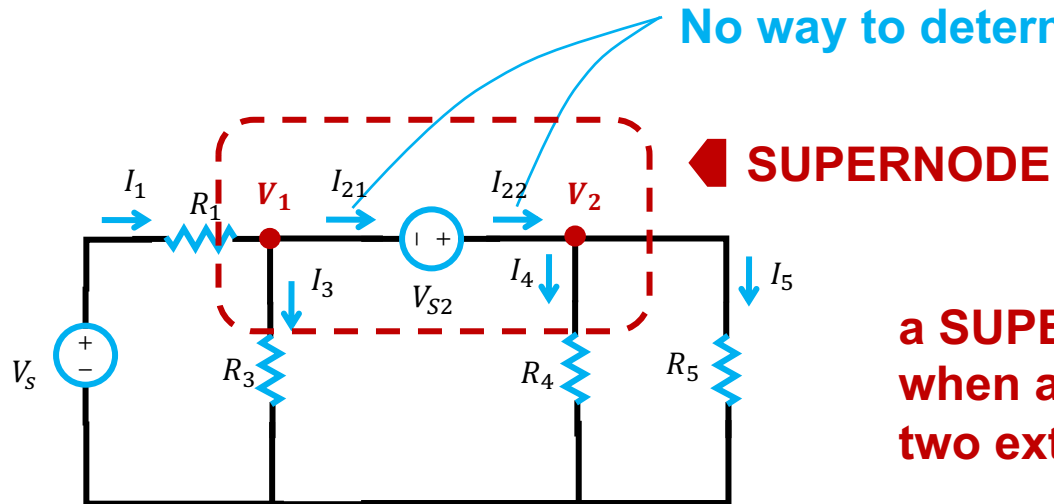
2 equations in 2 unknowns

* Note:

the # of equations to solve =
of extraordinary nodes – 1

Since we don't write an equation
for GND node. It's redundant

Supernode



a **SUPERNODE** is formed when a voltage source connects two extraordinary nodes



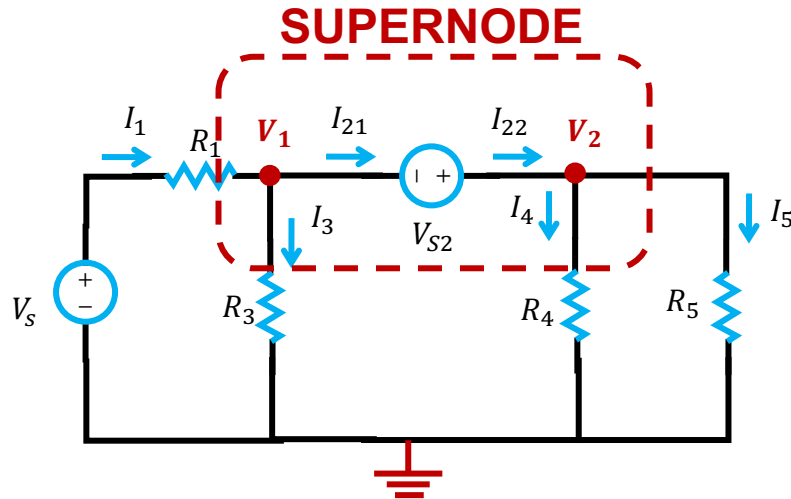
- According to KCL

$$I_1 - I_3 = I_{21} \quad I_{22} = I_4 + I_5$$

$$I_1 - I_3 - I_4 - I_5 = 0$$

The "internal" current (I_{21}, I_{22}) cancels

Supernode w/ node-voltage method



- Step 1a: identify all extraordinary nodes
- Step 1b: Assign **ground** node
- Step 1c: assign node voltages to the rest
- Step 2a: perform **KCL** at each node
- Step 2b: apply ohm's law for each current

@ the supernode

$$I_1 - I_3 - I_4 - I_5 = 0$$

$$\Rightarrow \frac{V_s - V_1}{R_1} - \frac{V_1}{R_3} - \frac{V_2}{R_4} - \frac{V_2}{R_5} = 0$$

1 equations in 2 unknowns

- Step 2c: generate an auxiliary equation at the supernode

$$V_2 - V_1 = V_{s2}$$

2 equations in 2 unknowns

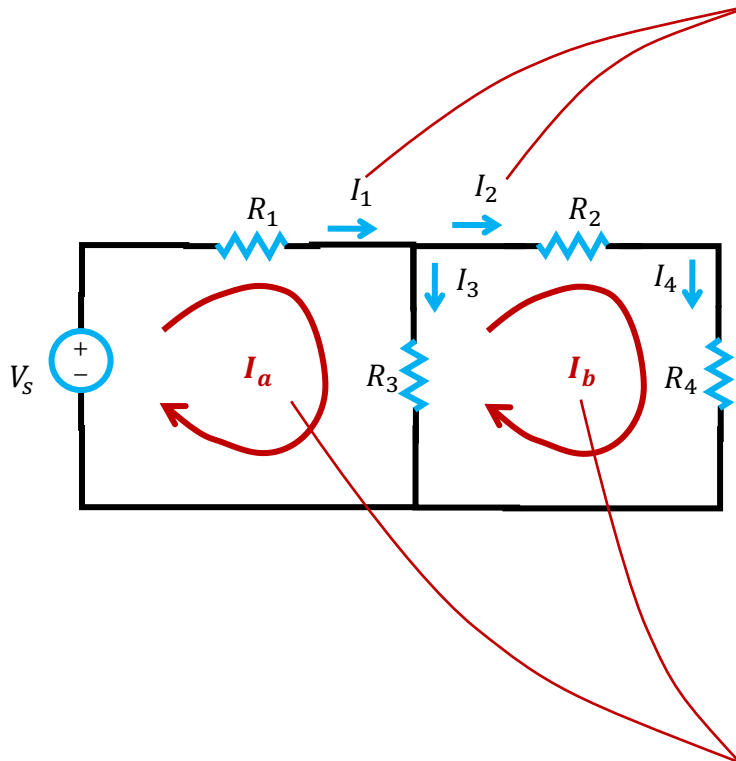
Outlines

- Node-voltage method
 - Basis of node voltage
 - Node-voltage method for circuit analysis
 - Supernode w/ node-voltage method

Outlines

- Node-voltage method
 - Basis of node voltage
 - Node-voltage method for circuit analysis
 - Supernode w/ node-voltage method
- Mesh-current method

Basis of mesh-current



Actual currents

- Case 1: mesh-current equals actual current

$$I_1 = I_a$$

* Note:

I_a is positive as it agrees with I_1 's direction

- Case 2: an element is shared by 2 meshes

$$I_3 = I_a - I_b$$

* Note:

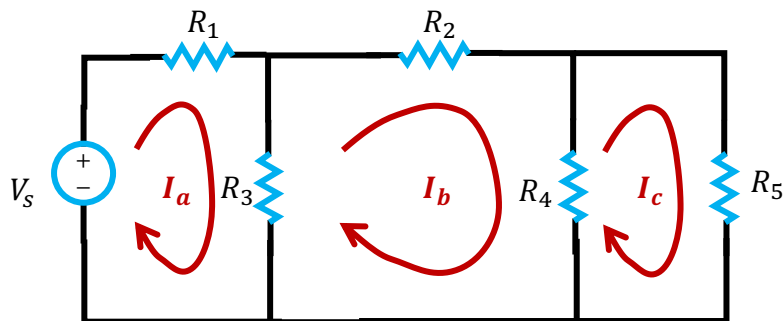
I_a is positive as it agrees with I_3 's direction

I_b is negative as it flows in opposite to I_3 's direction

Mesh-currents

Mesh-current method

QUESTION: Find the output current of the voltage source



- Step 1a: identify all mesh
- Step 1b: assign each mesh an unknown current, usually in clockwise direction

- Step 2a: apply KVL to each mesh

In mesh a

$$-V_s + I_a R_1 + (I_a - I_b) R_3 = 0$$

In mesh b

$$(I_b - I_a) R_3 + I_b R_2 + (I_b - I_c) R_4 = 0$$

In mesh c

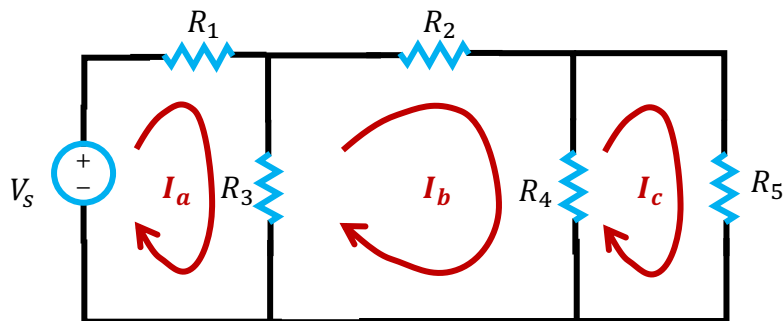
$$(I_c - I_b) R_4 + I_c R_5 = 0$$

* Note:

- Put all voltages in terms of mesh-current
- Each term is just Ohm's law
- Sign of the current

Mesh-current method

QUESTION: Find the output current of the voltage source



- Step 1a: identify all mesh
- Step 1b: assign each mesh an unknown current, usually in clockwise direction
- Step 2a: apply KVL to each mesh
- Step 2b: group terms by mesh-current

In mesh a

$$-V_s + I_a R_1 + (I_a - I_b) R_3 = 0$$

$$\blacksquare I_a(R_1 + R_3) - I_b R_3 = V_s$$

In mesh b

$$(I_b - I_a) R_3 + I_b R_2 + (I_b - I_c) R_4 = 0$$

$$\blacksquare -I_a R_3 + I_b(R_3 + R_2 + R_4) - I_c R_4 = 0$$

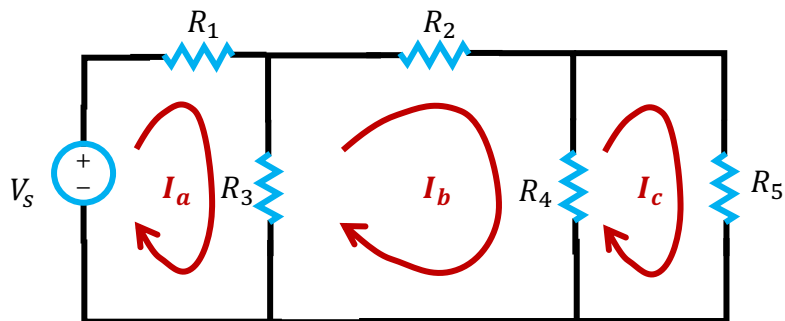
In mesh c

$$(I_c - I_b) R_4 + I_c R_5 = 0$$

$$\blacksquare -I_b R_4 + I_c(R_4 + R_5) = 0$$

Mesh-current method

QUESTION: Find the output current of the voltage source



$$\begin{cases} I_a(R_1 + R_3) - I_b R_3 = V_s \\ -I_a R_3 + I_b(R_3 + R_2 + R_4) - I_c R_4 = 0 \\ -I_b R_4 + I_c(R_4 + R_5) = 0 \end{cases}$$

- Step 1a: identify all mesh
- Step 1b: assign each mesh an unknown current, usually in clockwise direction
- Step 2a: apply KVL to each mesh
- Step 2b: group terms by mesh-current
- Step 3: solve the system of equations

3 equations in 3 unknowns

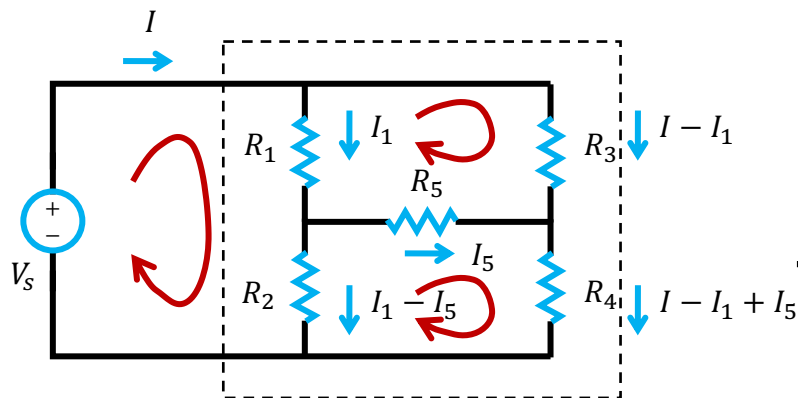
* Note:

All the actual currents can be calculated from the mesh-current, such as

$$I_{R_3} = I_a - I_b$$

Example 1 Wheatstone Bridge

QUESTION: Find the equivalent resistance of the circuit in the block



According to mesh-current method

$$\begin{cases} -V_s + R_1 I_1 + (I_1 - I_5)R_2 = 0 \\ R_1 I_1 + R_5 I_5 - (I - I_1)R_3 = 0 \\ -(I_1 - I_5)R_2 + R_5 I_5 + (I - I_1 + I_5)R_4 = 0 \end{cases}$$

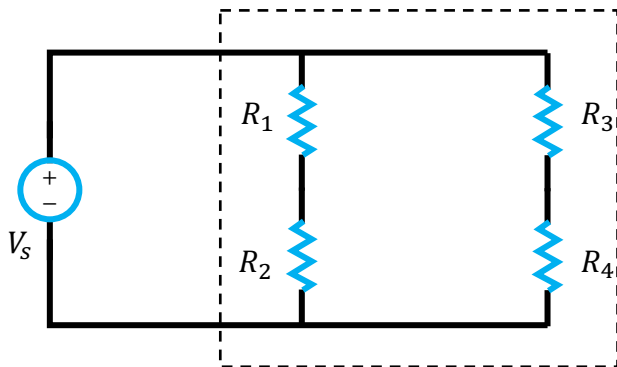
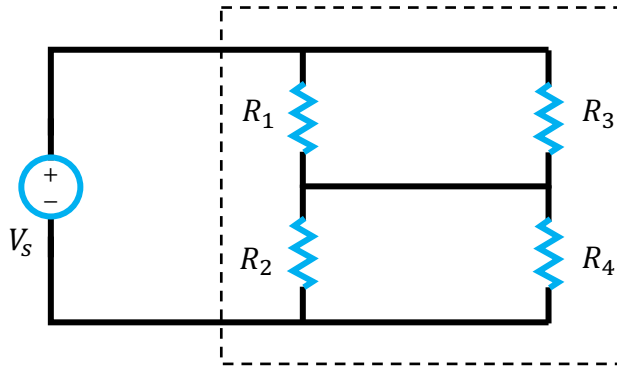
3 equations in 3 unknowns

$$\Rightarrow \begin{bmatrix} 0 & R_1 + R_2 & -R_2 \\ -R_3 & R_1 + R_3 & R_5 \\ -R_4 & R_2 + R_4 & -R_2 - R_4 - R_5 \end{bmatrix} \begin{bmatrix} I \\ I_1 \\ I_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} V_s$$

$$\Rightarrow R_{eq} = \frac{V_s}{I} = \frac{R_5(R_1 + R_2)(R_3 + R_4) + R_2 R_4(R_1 + R_3) + R_1 R_3(R_2 + R_4)}{R_5(R_1 + R_2 + R_3 + R_4) + (R_2 + R_4)(R_1 + R_3)}$$

Example 1 Wheatstone Bridge

$$R_{eq} = \frac{V_s}{I} = \frac{R_5(R_1 + R_2)(R_3 + R_4) + R_2R_4(R_1 + R_3) + R_1R_3(R_2 + R_4)}{R_5(R_1 + R_2 + R_3 + R_4) + (R_2 + R_4)(R_1 + R_3)}$$



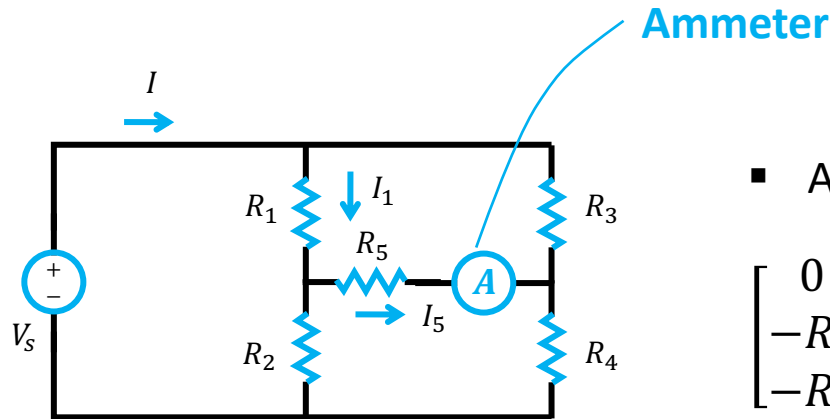
- If $R_5 = 0$ ➡ short circuit

$$\begin{aligned} R_{eq} &= \frac{R_2R_4(R_1 + R_3) + R_1R_3(R_2 + R_4)}{(R_2 + R_4)(R_1 + R_3)} \\ &= \frac{R_2R_4}{R_2 + R_4} + \frac{R_1R_3}{R_1 + R_3} \\ &= R_2 || R_4 + R_1 || R_3 \end{aligned}$$

- If $R_5 = \infty$ ➡ open circuit

$$\begin{aligned} R_{eq} &= \frac{(R_1 + R_2)(R_3 + R_4)}{R_1 + R_2 + R_3 + R_4} \\ &= (R_1 + R_2) || (R_3 + R_4) \end{aligned}$$

Example 1 Wheatstone Bridge



- According to system of equations

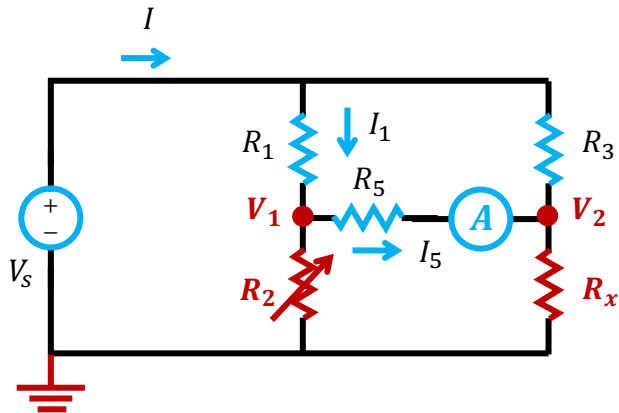
$$\begin{bmatrix} 0 & R_1 + R_2 & -R_2 \\ -R_3 & R_1 + R_3 & R_5 \\ -R_4 & R_2 + R_4 & -R_2 - R_4 - R_5 \end{bmatrix} \begin{bmatrix} I \\ I_1 \\ I_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} V_s$$

$$\blacktriangleright I_5 = \frac{R_2 R_3 - R_1 R_4}{R_5 (R_1 + R_2)(R_3 + R_4) + R_2 R_4 (R_1 + R_3) + R_1 R_3 (R_2 + R_4)} V_s$$

- Balanced condition: $I_5 = 0$

$$\blacktriangleright R_2 R_3 = R_1 R_4$$

Example 1 Wheatstone Bridge



▪ Balanced condition: $I_5 = 0$ $\Rightarrow V_1 = V_2$

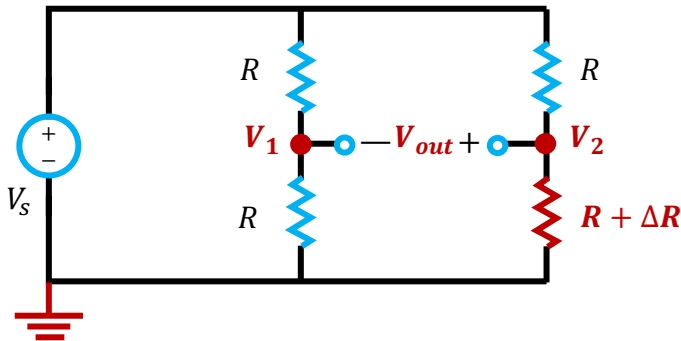
▪ According to voltage division

$$\begin{cases} V_1 = \frac{R_2}{R_1 + R_2} V_s \\ V_2 = \frac{R_x}{R_3 + R_x} V_s \end{cases} \Rightarrow R_x = \frac{R_2 R_3}{R_1}$$

Wheatstone bridge can be used to determine unknown resistance based on “balanced” condition

Example 1 Wheatstone Bridge

QUESTION: Find the voltage V_{out}



- According to voltage division

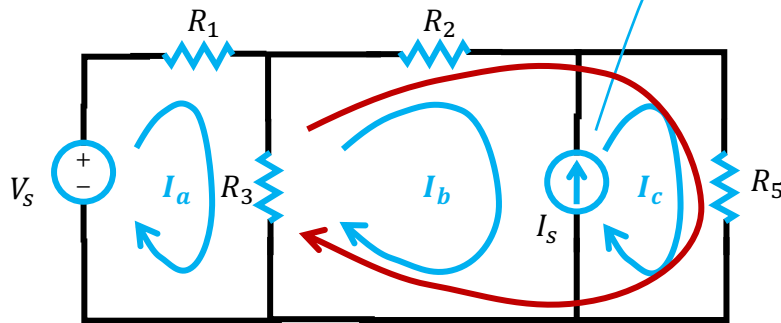
$$\begin{cases} V_1 = \frac{R}{R + R} V_s = \frac{1}{2} V_s \\ V_2 = \frac{(R + \Delta R)}{R + (R + \Delta R)} V_s = \frac{R + \Delta R}{2R + \Delta R} V_s \end{cases}$$

$$V_{out} = V_2 - V_1 = \frac{R + \Delta R}{2R + \Delta R} V_s - \frac{1}{2} V_s = \frac{\Delta R}{4R + 2\Delta R} V_s = \frac{\Delta R}{4R \left(1 + \frac{\Delta R}{2R}\right)} V_s \approx \frac{\Delta R}{4R} V_s$$

Since $\frac{\Delta R}{R} \ll 1$

In Wheatstone bridge, by measuring v_{out} , we can find ΔR

Supermesh



No way to determine this voltage!!

In mesh a

$$-V_s + I_a R_1 + (I_a - I_b) R_3 = 0$$

$$\blacksquare I_a (R_1 + R_3) - I_b R_3 = V_s$$

In the supermesh

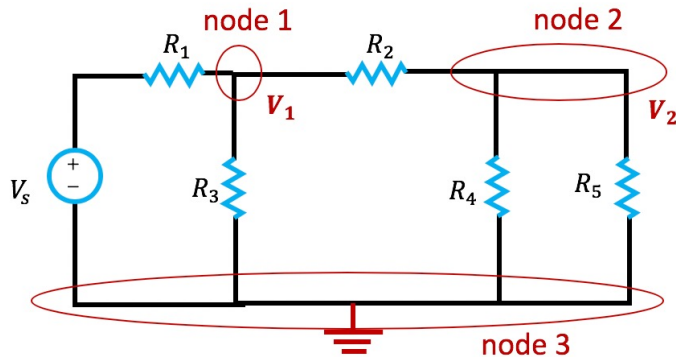
$$(I_b - I_a) R_3 + I_b R_2 + I_c R_5 = 0$$

Auxiliary equation for the supermesh

$$I_c - I_b = I_s$$

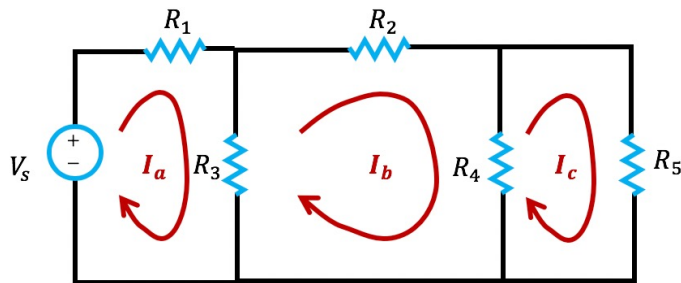
- Step 1b: assign each mesh an unknown current, usually in clockwise direction
- Step 2a: apply KVL to each mesh

Nodal versus Mesh



Node-voltage method

- Node voltages are UNKNOWN
- # of equations =
of extraordinary node - 1



Mesh-current method

- Currents flowing in meshes are UNKNOWN
- # of equations = # of meshes

Outlines

- Node-voltage method
 - Basis of node voltage
 - Node-voltage method for circuit analysis
 - Supernode w/ node-voltage method
- Mesh-current method
 - Basis of mesh-current
 - Mesh-current method for circuit analysis
 - Supermesh w/ mesh-current method
 - Node-voltage v.s. mesh-current method
- Linearity & Superposition

Recall: linear function

A function is linear if it satisfies 2 properties:

- **Additivity** (superposition) $f(x + y) = f(x) + f(y)$
- **Homogeneity** (scaling) $f(ax) = a f(x)$

A linear function therefore must exhibit both these 2 properties simultaneously:

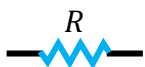
$$f(ax + by) = af(x) + bf(y)$$

Linearity describes a relationship between cause & effect

Linear device

A DEVICE is linear if it satisfies 2 properties:

- **Additivity**: response to SUM OF INPUTS is the SUM OF THE RESPONSES to each input applied separately



- According to Ohm's Law, if $v_1 = i_1 R$ and $v_2 = i_2 R$
- If a current $i = i_1 + i_2$ is applied to R

$$v = (i_1 + i_2)R = i_1 R + i_2 R = v_1 + v_2$$

- **Homogeneity**: if input (excitation) is multiplied by a constant, then output (response) is multiplied by the SAME constant



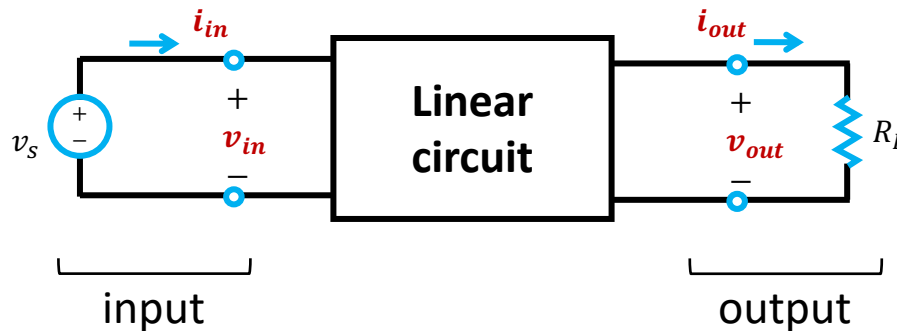
- According to Ohm's Law, if $v_1 = i_1 R$
- If the current is increased by k , $i_2 = k i_1$

$$v_2 = i_2 R = k i_1 R = k v_1$$

A resistor is a linear device since its voltage-current relationship satisfies both the additivity and homogeneity properties

Linear circuit

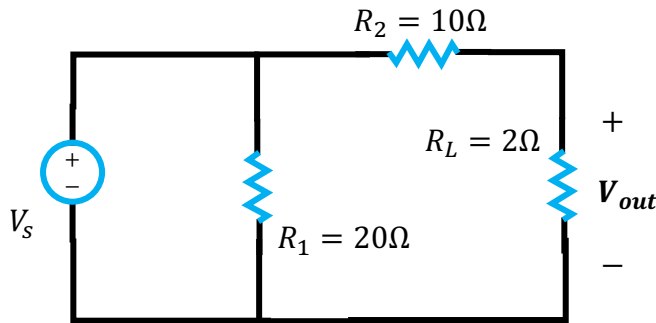
A CIRCUIT is linear if it possesses the properties of additive and homogeneity



- A circuit is linear when its output is directly proportional to its input
- A linear circuit's output must scale with its input
- In general: a linear circuit consists of only
 - ❑ linear devices (resistors, capacitors, inductors)
 - ❑ linear dependent sources
 - ❑ independent sources

Example 2

QUESTION: if $V_s = 24V$, Find the output voltage of the load R_L



- Assume $V_{out-test} = 1V$
- According to Ohm's law

$$I_{R_L} = \frac{V_{out-test}}{R_L} = 0.5A$$

$$\begin{aligned} V_{s-test} &= V_{R_2} + V_{out-test} \\ &= I_{R_L} R_2 + V_{out-test} = 6V \end{aligned}$$

- This is a linear circuit

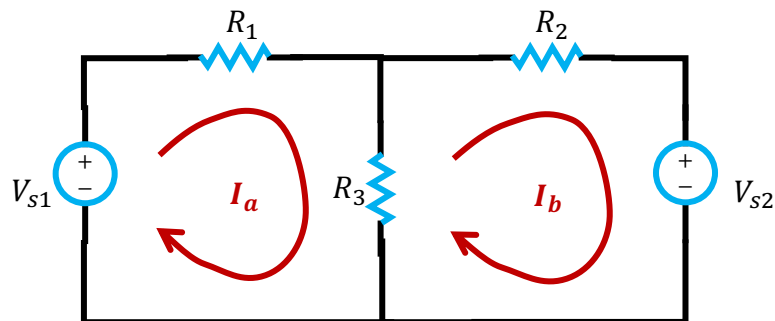
$$V_s = 4V_{s-test} \quad \blacktriangleright \quad V_{out} = 4V$$

Outlines

- Node-voltage method
 - Basis of node voltage
 - Node-voltage method for circuit analysis
 - Supernode w/ node-voltage method
- Mesh-current method
 - Basis of mesh-current
 - Mesh-current method for circuit analysis
 - Supermesh w/ mesh-current method
 - Node-voltage v.s. mesh-current method
- Linearity & Superposition
 - What is a linear circuit
 - Additivity & homogeneity of linear circuit
 - Homogeneity in circuit analysis
 - **Superposition in circuit analysis**

Example 3

QUESTION: Find the voltage on R_3



- According to mesh-current method

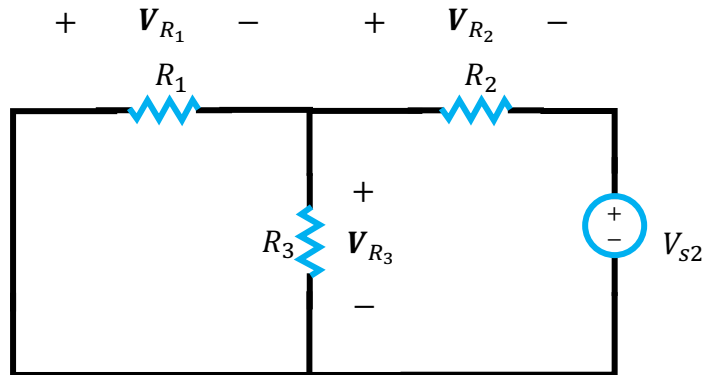
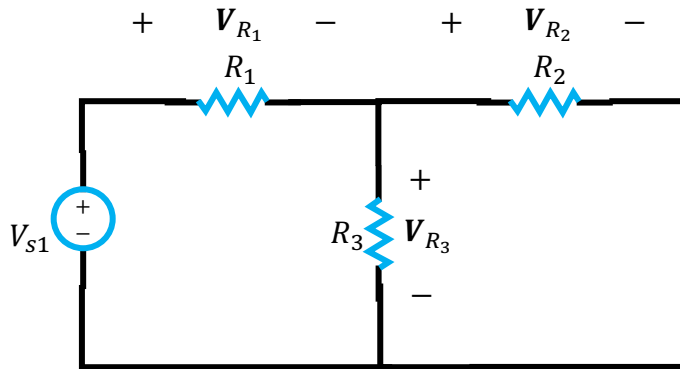
$$\begin{cases} -V_{s1} + R_1 I_a + R_3(I_a - I_b) = 0 \\ R_3(I_b - I_a) + R_2 I_b + V_{s2} = 0 \end{cases}$$

$$\Rightarrow \begin{cases} I_a = \frac{(R_2 + R_3)V_{s1} - R_3 V_{s2}}{R_1 R_2 + R_1 R_3 + R_2 R_3} \\ I_b = \frac{R_3 V_{s1} - (R_1 + R_3)V_{s2}}{R_1 R_2 + R_1 R_3 + R_2 R_3} \end{cases}$$

$$\Rightarrow V_{R_3} = R_3(I_a - I_b)$$

$$= \frac{R_2 R_3 V_{s1} + R_1 R_3 V_{s2}}{R_1 R_2 + R_1 R_3 + R_2 R_3}$$

Example 3



- According to voltage division

$$V'_{R3} = \frac{R_2 || R_3}{R_1 + R_2 || R_3} V_{s1}$$

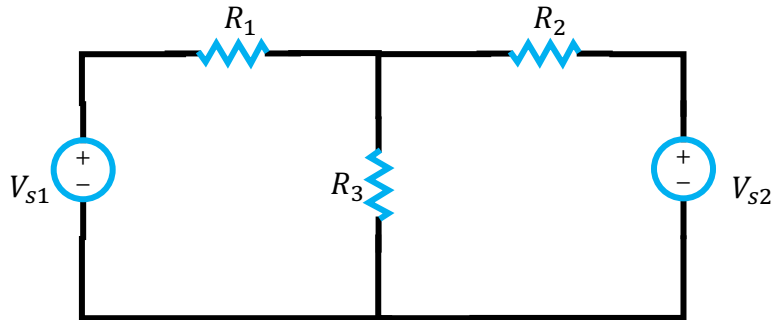
$$= \frac{R_2 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} V_{s1}$$

$$V''_{R3} = \frac{R_1 || R_3}{R_2 + R_1 || R_3} V_{s2}$$

$$= \frac{R_1 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} V_{s2}$$

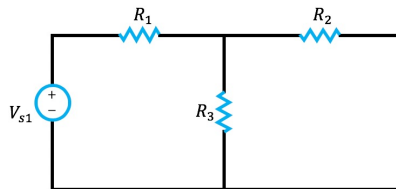
Example 3

QUESTION: Find the voltage on R_3



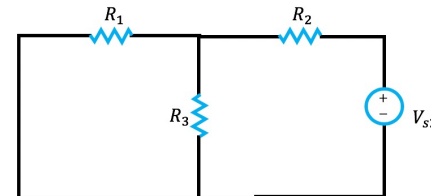
$$V_{R_3} = \frac{R_2 R_3 V_{s1} + R_1 R_3 V_{s2}}{R_1 R_2 + R_1 R_3 + R_2 R_3}$$

=



$$V'_{R_3} = \frac{R_2 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} V_{s1}$$

+



$$V''_{R_3} = \frac{R_1 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} V_{s2}$$

The superposition property of a linear circuit can help analyze circuits with more than 1 source

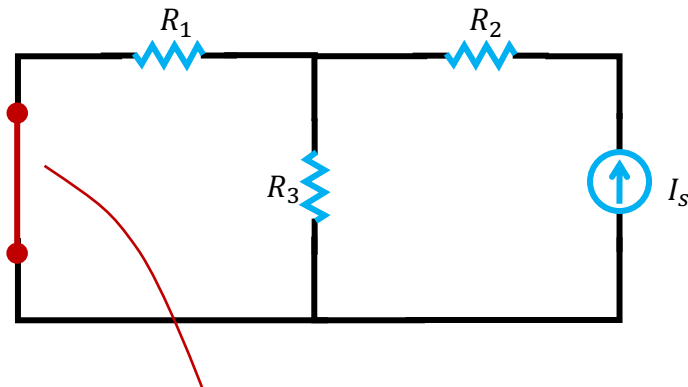
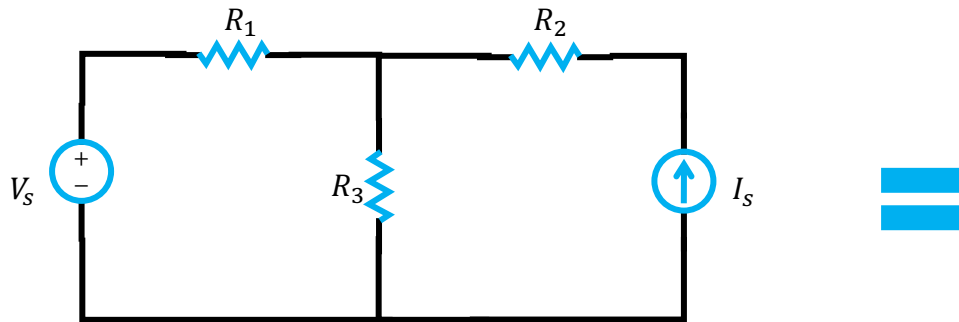
Superposition in Action

Superposition trades off the examination of several simpler circuits in place of one complex circuit

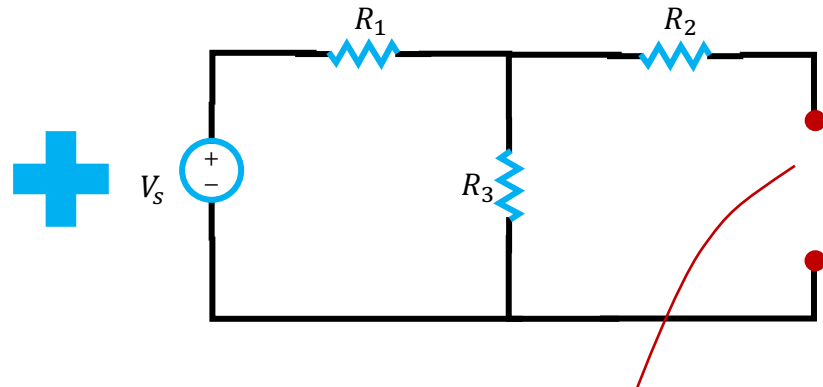
How to apply superposition to circuit analysis

- Consider one **INDEPENDENT** source at a time, by “turning off” all other **INDEPENDENT** sources
- **DEPENDENT** sources are left intact because they are controlled by circuit variables

How to “turn off” an independent src



**Short circuit –
Turn off voltage source**

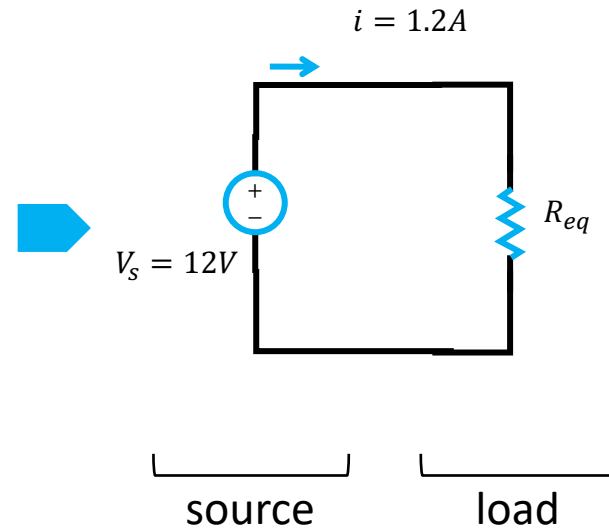
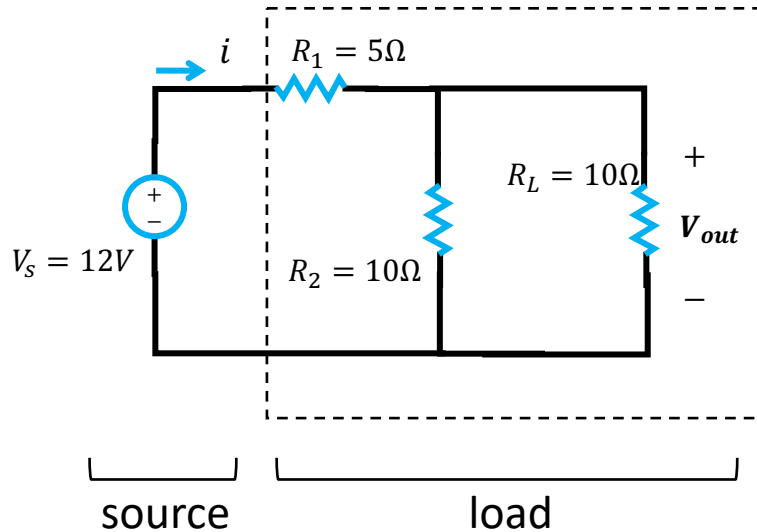


**Open circuit –
Turn off current source**

Outlines

- Node-voltage method
 - Basis of node voltage
 - Node-voltage method for circuit analysis
 - Supernode w/ node-voltage method
- Mesh-current method
 - Basis of mesh-current
 - Mesh-current method for circuit analysis
 - Supermesh w/ mesh-current method
 - Node-voltage v.s. mesh-current method
- Linearity & Superposition
 - What is a linear circuit
 - Additivity & homogeneity of linear circuit
 - Homogeneity in circuit analysis
 - Superposition in circuit analysis
- Thévenin and Norton Equivalent Circuits

Recall: equivalent resistance

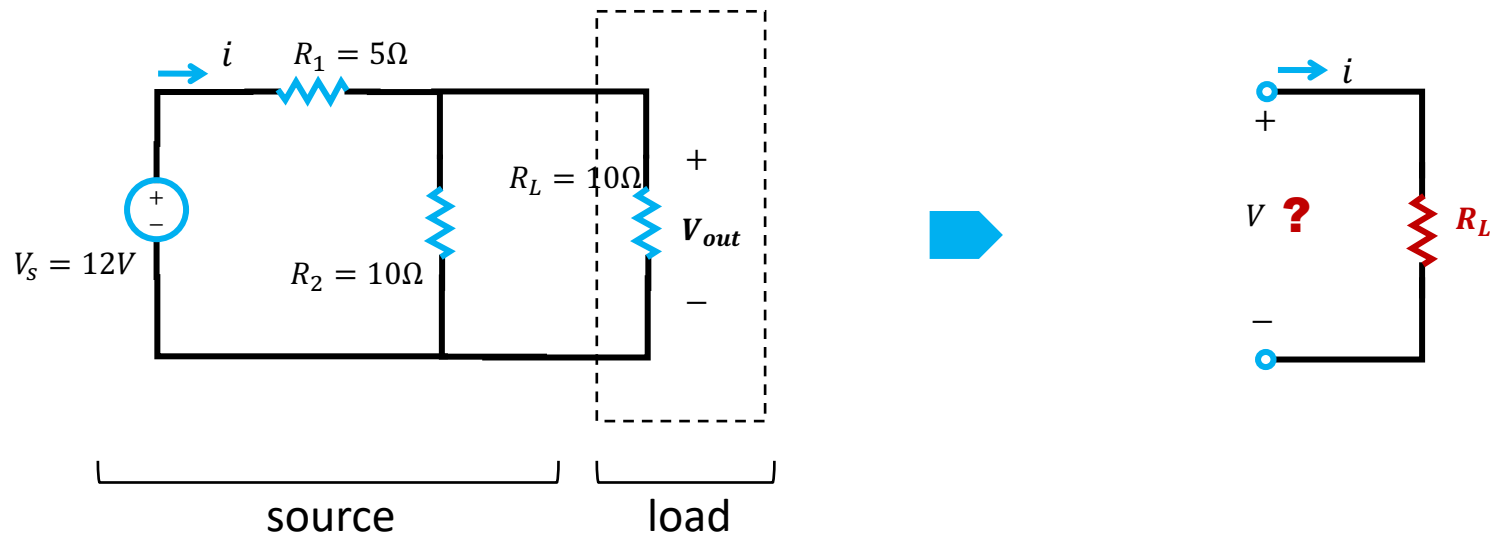


$$R_{eq} = R_1 + (R_2 || R_L) = 10\Omega$$

$$i = \frac{V_s}{R_{eq}} = 1.2A$$

- Source sees the load as just “resistance”
- Source supplies the amount of current demanded by load

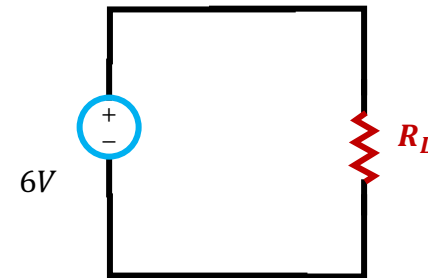
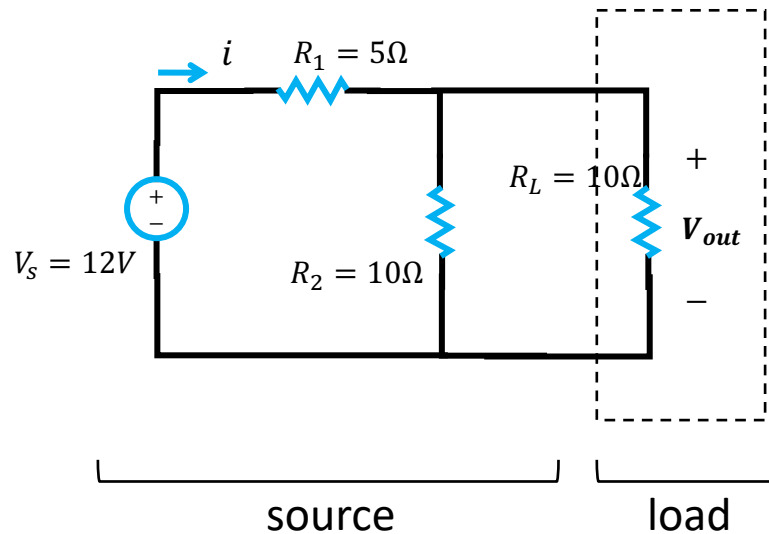
What does R_L see?



$$V_{R_L} = V_s - iR_1 = 6V$$

$$i_{R_L} = \frac{V_{R_L}}{R_L} = 0.6A$$

What does R_L see?



Are these two circuit
equivalent?

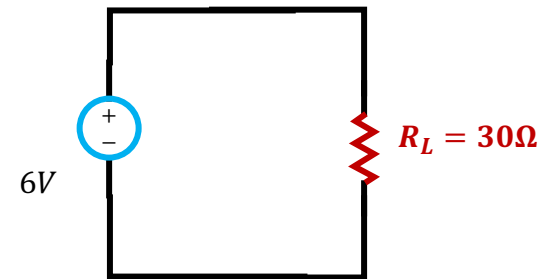
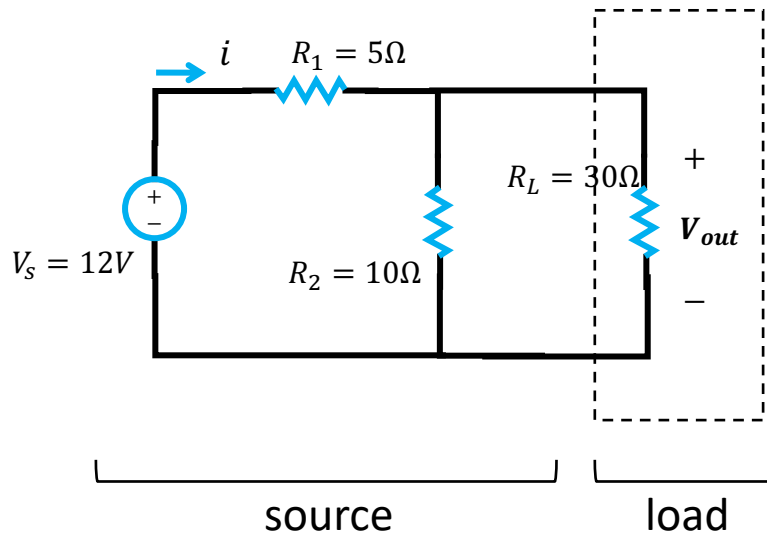
NO!!!

$$V_{R_L} = V_s - iR_1 = 6V$$

$$i_{R_L} = \frac{V_{R_L}}{R_L} = 0.6A$$

What does R_L see?

What if R_L increases to 30Ω ?



These two circuit are NOT equivalent!!!

$$V_{R_L} = V_S - iR_1 = 7.2V$$

$$i_{R_L} = \frac{V_{R_L}}{R_L} = 0.24A$$



$$i_{R_L} = \frac{V_S}{R_L} = 0.2A$$

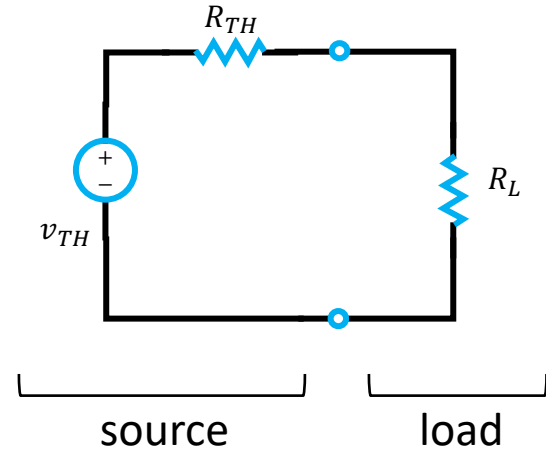
$$V_{R_L} = V_S = 6V$$

Equivalent circuits supply the same voltage & current to ANY load

Circuit equivalent

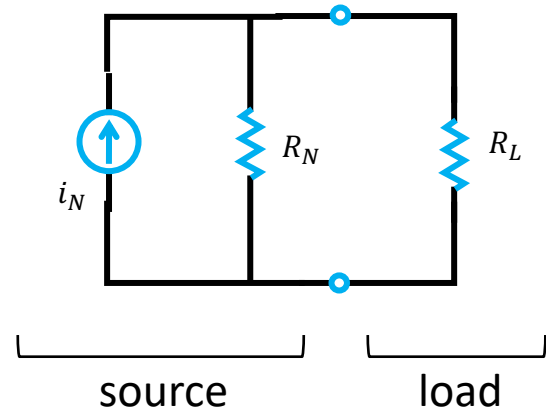
Thévenin's theorem

LINEAR two-terminal circuit can be replaced by an equivalent circuit composed of a voltage source and a series resistor



Norton's theorem

LINEAR two-terminal circuit can be replaced by an equivalent circuit composed of a current source and a parallel resistor

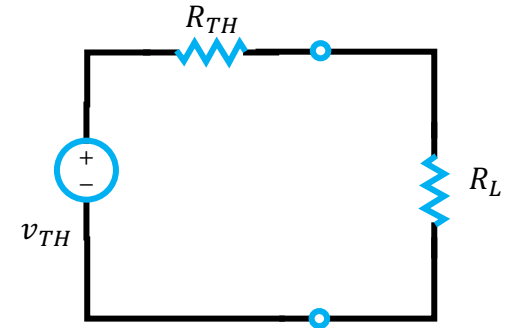


Thévenin & Norton equivalency

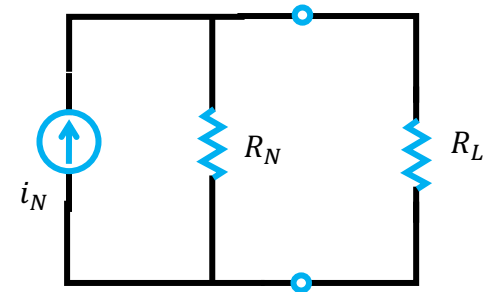
- According to source transformation

$$\begin{cases} i_N = \frac{v_{TH}}{R_{TH}} \\ R_{TH} = R_N \end{cases}$$

- Norton's Theorem is just a “source transformation” of Thévenin's theorem
- Superposition and Additivity make these theorems possible



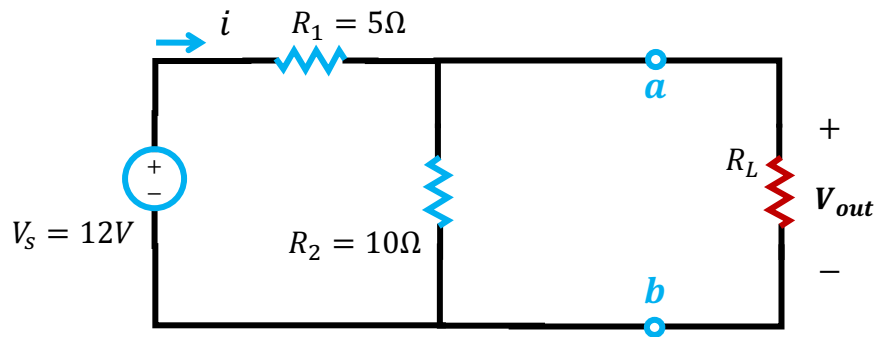
Thévenin's theorem



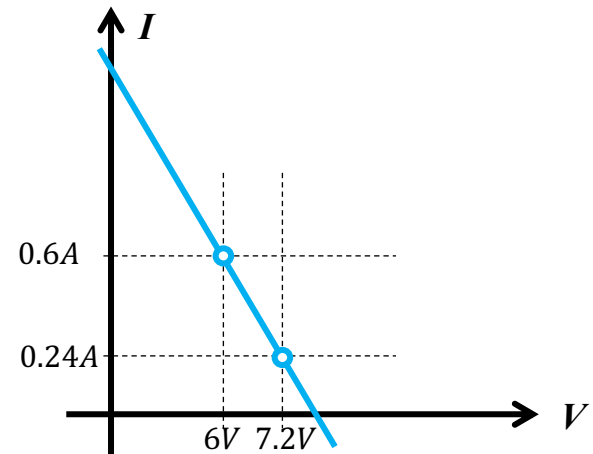
Norton's theorem

Example 4

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



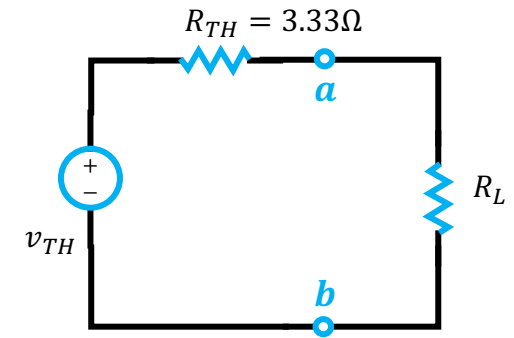
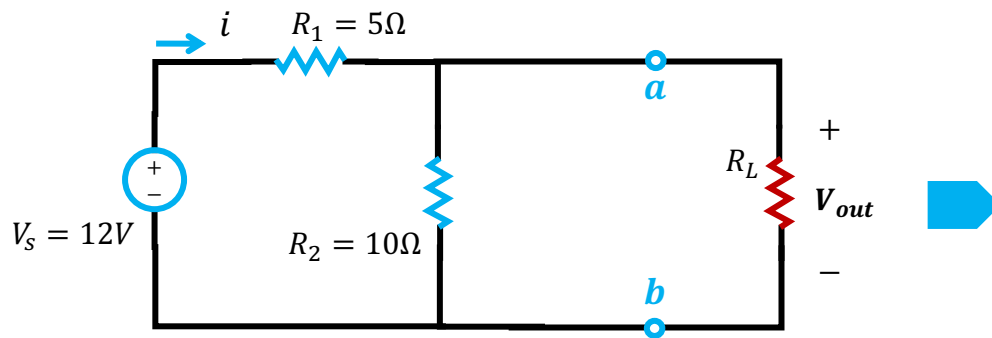
- When $R_L = 10\Omega$ $\left\{ \begin{array}{l} i_{R_L} = 0.6A \\ V_{R_L} = 6V \end{array} \right.$
- When $R_L = 30\Omega$ $\left\{ \begin{array}{l} i_{R_L} = 0.24A \\ V_{R_L} = 7.2V \end{array} \right.$



$$R_{TH} = \frac{\Delta V}{\Delta I} = \frac{7.2V - 6V}{0.6A - 0.24A} = 3.33\Omega$$

Example 4

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b

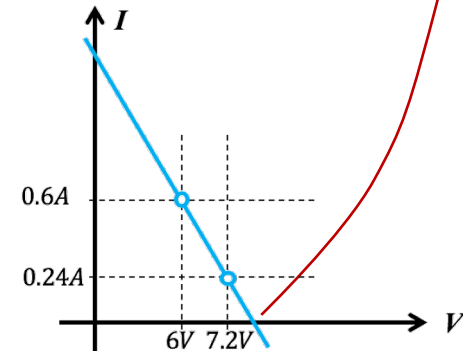


- When $R_L = 10\Omega$

$$\begin{cases} i_{R_L} = 0.6A \\ V_{R_L} = 6V \end{cases}$$
- When $R_L = 30\Omega$

$$\begin{cases} i_{R_L} = 0.24A \\ V_{R_L} = 7.2V \end{cases}$$

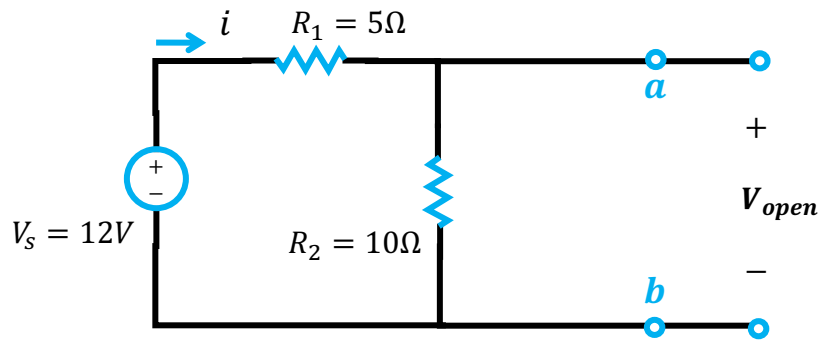
$$v_{TH} = (R_{TH} + R_L)i_{R_L} = 8V$$



v_{TH} is the maximum voltage @open circuit ($i = 0$)

Example 4

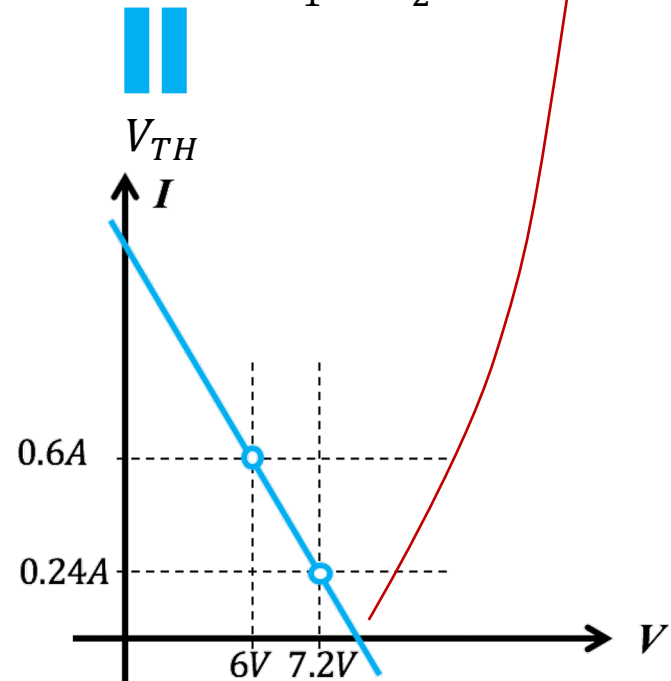
QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



- Step 1: remove the load
- Step 2: find V_{open}

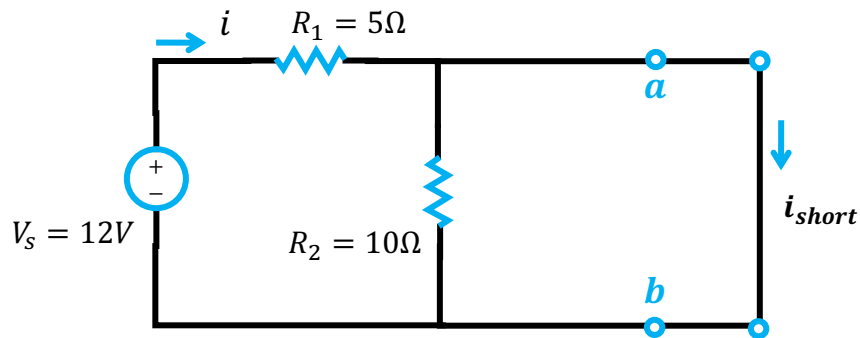
- According to voltage division

$$V_{open} = \frac{R_2}{R_1 + R_2} V_s = 8V$$



Example 4

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b

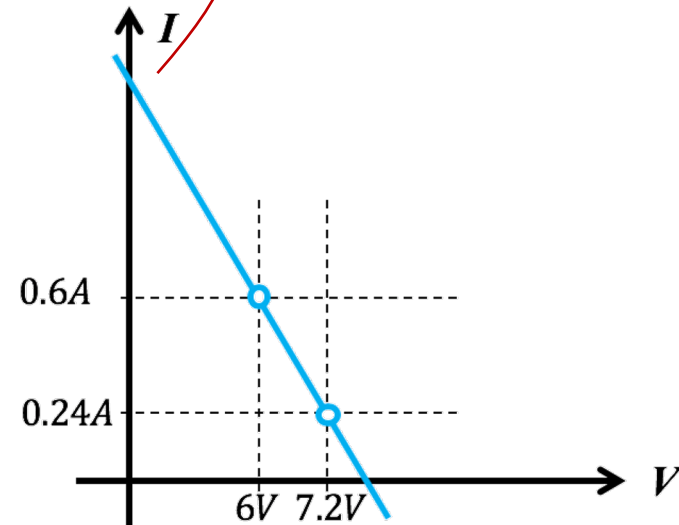


- Step 1: remove the load
- Step 2: find V_{open}
- Step 3: find i_{short}

- According to Ohm's law

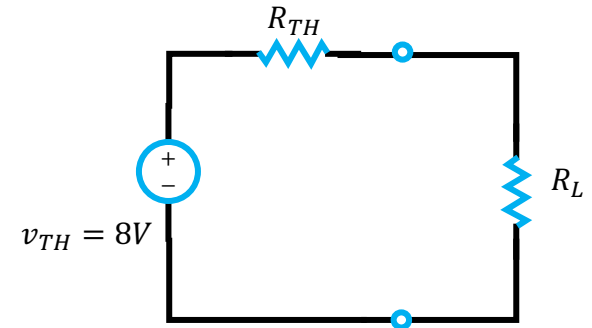
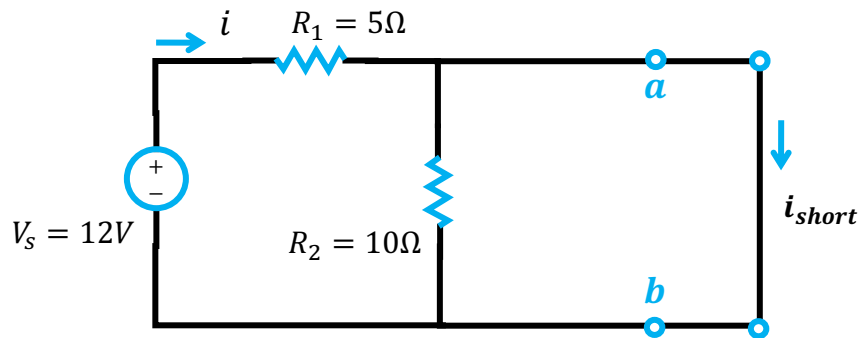
$$i_{open} = \frac{V_s}{R_1} = 2.4A$$

i_N



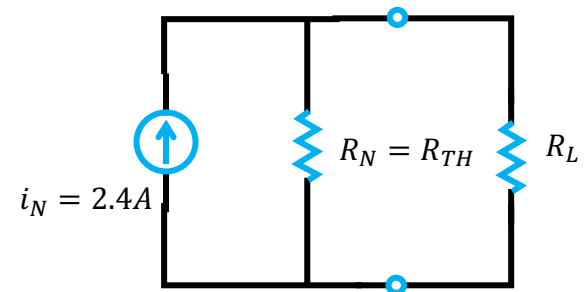
Example 4

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



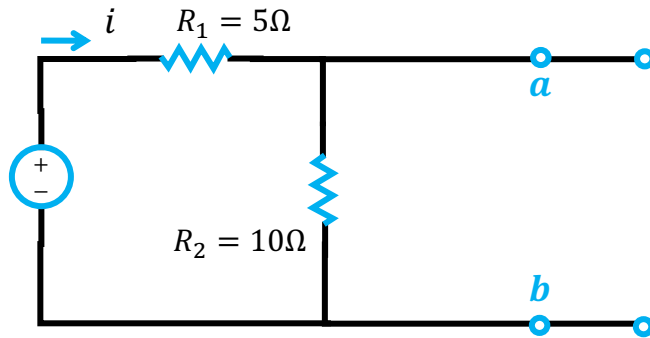
- Step 1: remove the load
- Step 2: find $V_{open} = v_{TH}$
- Step 3: find $i_{short} = i_N$
- Step 4: find R_{TH}

$$R_{TH} = \frac{v_{TH}}{i_N} = 3.33\Omega$$



Example 4

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



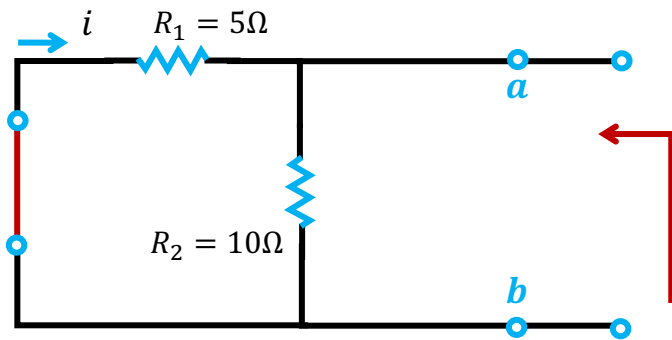
If a circuit contains ONLY independent sources

Method to find R_{TH}

- Step 1: remove the load

Example 4

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



R_{eq}

$$R_{TH} = R_{eq} = R_1 || R_2 = 3.33\Omega$$

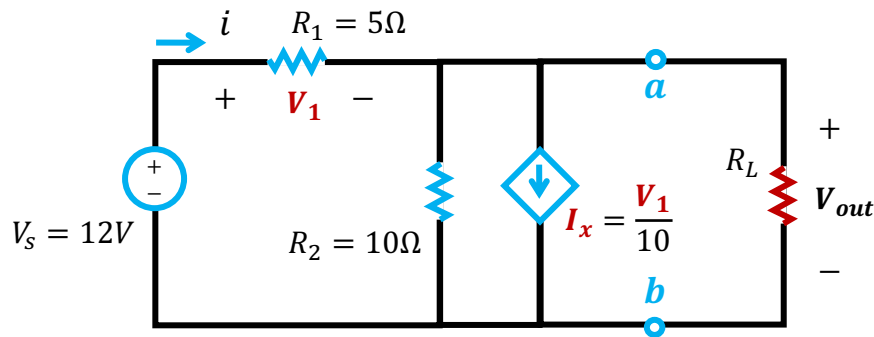
If a circuit contains ONLY independent sources

Method to find R_{TH}

- Step 1: remove the load
- Step 2: turn off ALL independent sources
 - SHORT all voltage sources
 - OPEN all current sources
- Step 3: find R_{eq} from the perspective of the load

Example 5

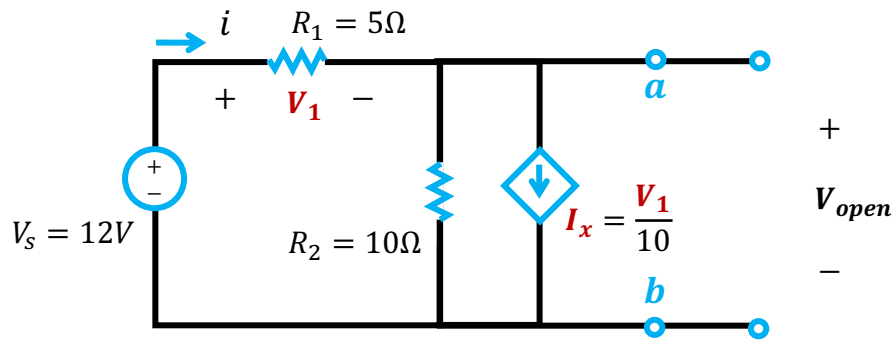
QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



**What if there is a
dependent source?**

Example 5

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



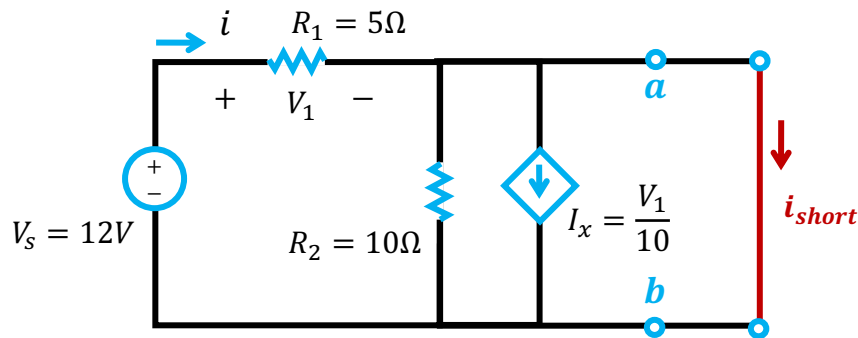
- Step 1: remove the load
- Step 2: find $V_{open} = v_{TH}$

$$\begin{cases} V_{open} = V_s - V_1 \\ \frac{V_1}{R_1} = \frac{V_{out}}{R_2} + I_x \end{cases}$$

➡ $V_{open} = 6V$

Example 5

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



- Step 1: remove the load
- Step 2: find $V_{open} = v_{TH}$

$$\begin{cases} V_{open} = V_s - V_1 \\ \frac{V_1}{R_1} = \frac{V_{out}}{R_2} + I_x \end{cases}$$

➡ $V_{open} = 6V$

- Step 3: find $i_{short} = i_N$
- Step 4: find R_{TH}

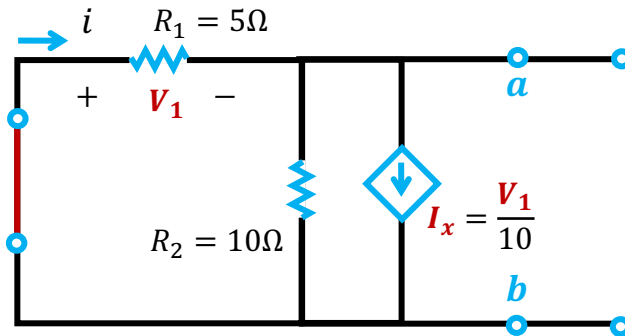
$$i_{short} = \frac{V_s}{R_1} - I_x$$

➡ $i_{short} = 1.2A$

$$R_{TH} = \frac{v_{TH}}{i_N} = 5\Omega$$

Example 5

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



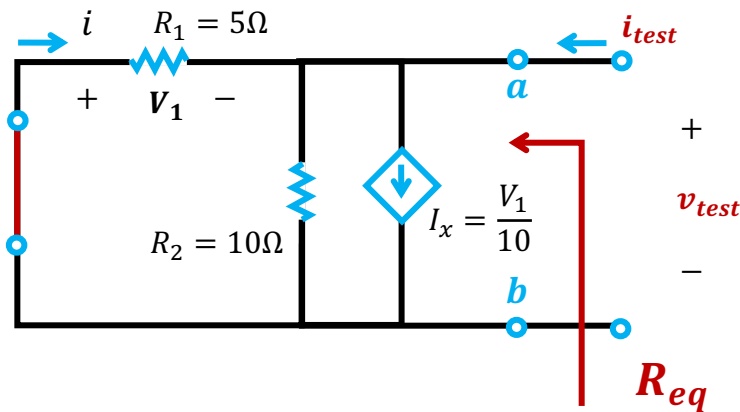
If a circuit contains DEPENDENT sources

Method to find R_{TH}

- Step 1: remove the load
- Step 2: turn off ALL independent sources
 - SHORT all voltage sources
 - OPEN all current sources
- Step 3: find R_{eq} from the perspective of the load

Example 5

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



- Step 3: find R_{eq} from the perspective of the load
 - Step 3.1: apply a testing voltage
 - Step 3.2: find the current generated by applying the testing voltage

$$i_{test} = \frac{v_{test}}{R_1} + \frac{v_{test}}{R_2} + I_x = \frac{v_{test}}{R_1} + \frac{v_{test}}{R_2} - \frac{v_{test}}{10}$$

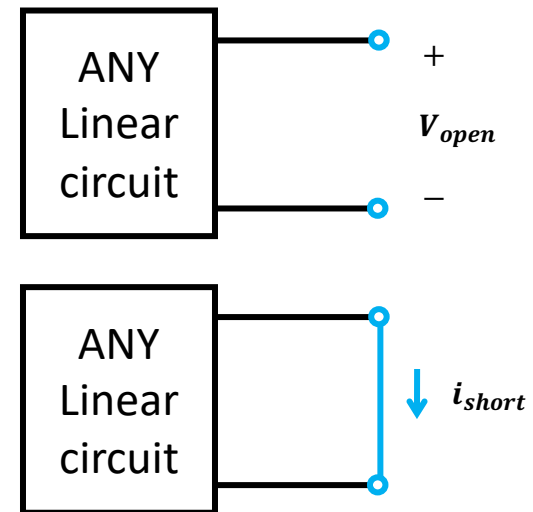
$$R_{TH} = R_{eq} = \frac{v_{test}}{i_{test}} = 5\Omega$$

Thévenin & Norton equivalency

How to find Thévenin / Norton equivalent circuit

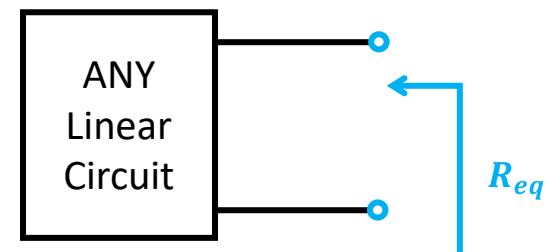
Method 1 – applied to ANY circuit

- Step 1: remove the load
- Step 2: $v_{TH} = V_{open}$
- Step 3: $i_N = i_{short}$
- Step 4: $R_{TH} = \frac{v_{TH}}{i_N}$



Method 2 – alternative way to find R_{TH} (equivalent resistance from the output)

- Step 1: remove the load
- Step 2: turn off ALL independent sources
 - SHORT all voltage sources
 - OPEN all current sources
- Step 3: find R_{eq} from the perspective of the load



Outlines

- Node-voltage method
 - Basis of node voltage
 - Node-voltage method for circuit analysis
 - Supernode w/ node-voltage method
- Mesh-current method
 - Basis of mesh-current
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 - Supermesh w/ mesh-current method
 - Node-voltage v.s. mesh-current method
- Linearity & Superposition
 - What is a linear circuit
 - Additivity & homogeneity of linear circuit
 - Homogeneity in circuit analysis
 - Superposition in circuit analysis
- Thévenin and Norton Equivalent Circuits
 - What is Thévenin and Norton Equivalency
 - How to calculate Thévenin and Norton Equivalent circuit

Reading tasks & learning goals

- Reading tasks

- Basic Engineering Circuit Analysis, 10th edition
 - Chapter 3, 5.1-5.3

- Learning goals

- Be able to calculate all V/I in circuits contain multi-nodes/loops
- Be able to use **node-voltage** method for circuit analysis
- Be able to use **mesh-current** method for circuit analysis
- Be able to calculate a **Thévenin equivalent** circuit for a linear circuit
- Be able to calculate a **Norton equivalent** circuit for a linear circuit